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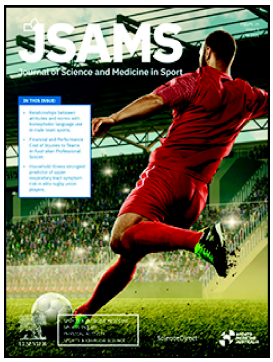

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Implicit doping attitude and athletes' accuracy in avoiding unintentional doping when being offered beverages with banned performance-enhancing substances

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Abstract

Objectives: Implicit doping attitude reflects the strength of one's automatic or unconscious evaluation of doping. This cross-societal study examined how implicit doping attitude predicted athletes' accuracy in avoiding unintentional doping.

Design: A real-time experimental design with cross-sectional data from three geographical regions.

Methods: A total of 681 athletes (M_{age} = 28.21, *SD* = 8.43; female = 47.1%) from Hong Kong (*N* = 177), Australia (*N* = 164) and the United Kingdom (N = 340) completed two real-time experimental tasks for measuring their implicit doping attitude (a brief single-category implicit association test) and the accuracy of avoiding unintentional doping (a novel canned beverage sorting task).

Results: Implicit doping attitude was significantly and negatively (β = -.09, $p < .05$, R^2 =.03, 95% CI of β = -1.64 to -.15) associated with athletes' accuracy in avoiding unintentional doping by screening out beverages with banned performance-enhancing substances. This association was maintained when we statistically controlled for the effects of society.

Conclusion: Athletes with a positive implicit doping attitude were less accurate in determining whether unknown beverages with the possible presence of banned performance-enhancing substances should be consumed. The negative association between implicit doping and athletes' accuracy in avoiding unintentional doping appeared to be consistent across societies.

Keywords: Drug use in sport; implicit association test; unintentional doping; avoidance of unintentional doping; safe sports behaviours.

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1. Introduction

Although the use of performance-enhancing drugs is strictly prohibited in sports, unintentional doping due to the unwitting ingestion of banned substances by means of food, supplements or medication could also violate the anti-doping codes of the World Anti-Doping Agency.[1, 2] According to a clinical review, most unintentional doping cases occur due to negligence and the careless use of dietary supplements or energy products.[3] The risk of unintentional doping is present largely because 10%-15%[4] of dietary supplements/medications on the market may contain banned performance-enhancing substances such as ephedrine, prohormones and anabolic steroids, which are not always correctly/clearly labelled.[5-7] The consumption of these products without careful self-checking or screening by athlete support personnel may substantially increase the risk of inadvertently ingesting banned performance-enhancing substances.[8] Consequences of such negligence may result in sanctions or defamation for athletes under the strict liability principle of the anti-doping rule. Specifically, this principle states that athletes are solely responsible for the presence of banned substances found in their doping tests regardless of whether it was used intentionally or unintentionally. Therefore, athletes must heighten their personal agency and awareness towards the avoidance of the accidental intake of prohibited substances, and remain extra vigilant when checking the ingredients of unfamiliar dietary/energy products before consumption. The purpose of this study is to determine the extent to which implicit doping attitude predicts athletes' accuracy of avoiding unintentional doping when presented with consumable items containing illicit substances.

Previous studies have demonstrated that athletes' various psychological factors such as motivations, trait self-control, and explicit doping attitude may explain athletes' behavioural patterns of the avoidance of unintentional doping.[1] However, these studies often evaluate psychological factors (e.g., attitude) and behavioural adherence using self-reported measures, which could confound the responses of participants due to social desirability or self-serving bias

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in this relatively sensitive topic associated with doping in sports.[9] Therefore, doping attitudes have shifted to using implicit association tests[10] to measure athletes' implicit attitude towards doping, and may provide a viable solution to overcome the limitations of self-reported measures.[11, 12]

Implicit attitudes reflects the strength of an individual's automatic or unconscious evaluation of a particular subject, as opposed to explicit attitudes, which involves more deliberative cognitive process[13]. These implicit attitudes are typically measured and indicated by a difference in reaction times when individuals complete an implicit association test. In this test, participants sort stimuli (i.e., doping-related words) into two categories (i.e., good vs. bad).[13, 14] The difference in salience between the categories of the task is then calculated (D-score) and the results may suggest the direction and relative strength of these automatic associations. It is therefore implied that the stronger the automatic association, the faster the reaction time. These results are different from traditional explicit attitude (i.e., measured by self-reported scales) as implicit attitude concerns about the extent to which the association between situational cues and behavioural actions is governed by unconscious, automatic, and deliberative processes.[15] It is therefore preferred for topics that confound with social desirability as these can be more difficult to mask. The theoretical basis of implicit attitude and implicit association tests are well established particularly within the literature of social psychology[16, 17] and health psychology.[18, 19] Since the introduction of the implicit association test in the literature[14], there has been a growing number of studies in sport science and medicine that examined athletes' implicit doping attitude.[11, 12, 20, 21] Athletes who hold positive implicit attitudes towards doping may indicate that they unconsciously possess favourable beliefs, attitudes, or feelings towards banned performance-enhancing substances.[21] Chan and colleagues[11] conducted the only study to date that examined the association between implicit doping attitude and athletes' behavioural responses of the avoidance of unintentional doping. In their study, they presented athletes with

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an unknown brand of lollipop, the researchers subsequently recorded whether the athlete ate the lollipop or not prior to reading the ingredients list. It was found that implicit doping attitude was negatively associated (r = -.23, p <.01) with athletes' reading of the ingredient list when they were given an unknown food product. However, implicit doping attitude surprisingly formed significant and positive associations (r = .18, p <.05) with self-reported behavioural adherence to the avoidance of unintentional doping and athletes' checking behaviour for the presence of banned performance-enhancing substances in the ingredient table.[11] The findings were mixed and did not provide a clear picture of how implicit doping attitude could explain athletes' behavioural patterns of avoiding unintentional doping. Moreover, there were some methodological limitations such as sample from one society, small sample size (n=143) and most importantly, the use of self-reported behavioural outcomes. Because of these limitations, we therefore, argue that the use of self-reported behaviour might have undermined the true association between implicit attitude and the avoidance of unintentional doping.

To address this knowledge gap, our study aims to test whether implicit doping attitude is associated with athletes' accuracy in screening out beverages with banned performance-enhancing substances (i.e., the accuracy of avoiding unintentional doping). Accuracy refers to their ability to correctly detect and subsequently sort ingredients that may or may not contain banned substances. We have hence devised a canned-beverage sorting task, allowing assessment of athlete's accuracy in the avoidance of unintentional doping. Based on previous research regarding the psychological factors of the avoidance of unintentional doping[11, 22], it was hypothesised that athletes' implicit doping attitude would be significantly and negatively associated with the accuracy of avoiding unintentional doping. We included a multi-societal sample involving athletes from Hong Kong (China), Australia, and the United Kingdom to examine the stability of the hypothesised association across societies, and we hypothesised that the negative association between implicit doping attitude and the accuracy of avoiding

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unintentional doping would hold when taking society, gender, age, education level, sport type, and sport level into account.

2. Methods

2.1. Participants

We recruited 681 athletes (M_{age} = 28.21, *SD* = 8.43; female = 47.1%, 1.3% did not reveal) from Hong Kong (*N* = 177), Australia (*N* = 164) and the United Kingdom (*N* = 340) who met our inclusion criteria of (1) aged 16 years or above and (2) participation in organised sports training and competitions at a regional level or above for 2 years or more. Statistical power analysis using G*Power (version 3.1.9.7) showed that a sample size of 649 is adequate for detecting a small effect size of .025 (estimated based on the average effect size of the predictive power of implicit doping attitude) and achieving the power of .85 (α = .05) for hierarchical linear regression analysis with 7 predictors. Therefore, our sample size was considered adequate.

Participants classified their competitive level as regional (*N* = 543), state (*N* = 69), national (*N* = 39) or international (*N* = 23); 7 participants did not disclose their competitive levels. Each participant's sport was also categorised into team (*N* = 369) or individual (*N* = 294) sports type, although 18 participants' sports could not be determined. Participants spent an average of 8.22 hours (*SD* = 6.46) training per week. In terms of education level, 97 participants completed secondary school, 121 completed post-school training college, 340 completed undergraduate, 121 completed postgraduate university, and 2 participants had undisclosed education backgrounds.

2.2. Procedures

This study was approved by the Institutional Review Board of [institution masked for blinded review]. Each participant (and their parent or legal guardian if under the age of 18)

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signed an informed consent form prior to the study. Participants were then asked to complete a brief survey about their demographic variables (e.g., age, gender, education level, sport type, sport level) before they performed the real-time experimental tasks for measuring their implicit doping attitude and accuracy of avoiding unintentional doping. A trained research associate provided instruction and guidance on the installation and execution of the smartphone application that ran the real-time experimental tasks. Participants received approximately £5 as reimbursement for their time contributions to the study.

2.3. Measures

Following the protocol developed by Chan and colleagues[11], we used the smartphone version of the brief single-category implicit association test[21] to assess implicit doping attitude. In this task, participants were asked to respond to the stimuli of words related to “doping” (i.e., steroid, narcotics, stimulants, diuretics), “I like” (i.e., pleasure, love, freedom, happy), and “I dislike” (i.e., evil, stink, filth, crash) as quickly and accurately as possible by pressing the left or the right button on the screens of the smartphones. Block A was the “doping – I like” block where participants pressed the right button for the words of “I like” and “doping”, and the left button for the words of “I dislike”. Block B was the “doping – I dislike” block where participants pressed the right button for the words of “I dislike” and “doping”, and the left button for the words of “I like”. Participants were randomly assigned block order. A practice trial was given to the participants prior to each block, so there was a total of 4 trials in the testing procedure. Error responses were accounted for and responses under 350ms and beyond 3,000ms were treated as outliers that were not indicative of automatic implicit attitude. Each participant’s implicit doping attitude was captured by their D-score, which is calculated by dividing the difference in the average reaction times of correct responses between blocks B and A by the standard deviation of the reaction times of correct responses in both blocks.[11] A positive implicit doping

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attitude would be indicated by a higher positive D-score, representing a positive or favourable implicit attitude towards doping.[20, 23]

To evaluate the accuracy of avoiding unintentional doping, we used a novel canned beverage sorting task that simulated the scenario of screening out banned performance-enhancing substances when consuming food products in athletes' everyday lives. At the start of the task, participants were given instructions on how to sort and identify canned beverages based on whether they contained banned substances. Specifically, beverages without an ingredients list or those containing banned substances are not to be consumed. To indicate their answers, they were instructed to press "can drink" (beverages with an ingredient list that does not contain banned performance-enhancing substances) and "cannot drink" (for other beverages that contained banned performance enhancing substances or those without an ingredient list).

Once participants understood the instructions, they clicked the "start practice trial" button allowing them to practice sorting the canned beverages before proceeding to the main task. In the beginning of both the practice trial and main task, a pop-up window will appear and ask whether participants are in a comfortable environment and have adequate focus for the following task. After the participants click yes, the task will begin.

In the task, participants were presented with one canned-beverage in the middle of the screen and were asked to respond by pressing the buttons labelled "can drink" or "cannot drink". Through a randomised order presentation of 32 images, the participants were presented with either (1) a canned beverage with a single banned substance on the ingredient list (i.e., steroids, stimulants, human growth hormones, narcotics, erythropoietin) or (2) a canned beverage without an ingredients list, or (3) a canned beverage with an ingredient list that that did not contain banned substances (see Appendix 1 for examples). To help participants inspect the ingredients list, they were also given a zoom function via the pinching motion.

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Consistent with the stimuli and response sequence of implicit association tests[14], the next image would appear when participants submitted a correct response; otherwise, an instant feedback labelled “wrong” would be presented before showing the next image. The order of presentation was randomised and the same 32 canned beverages were presented to all participants in the task. The task was scored based on the sum of correct responses out of the 32 trials. A higher total score represented greater accuracy in participants correctly sorting the beverage based on its ingredients lists.

2.4. Data analysis

We used hierarchical linear regression to examine our hypotheses regarding the association between implicit doping attitude and the accuracy of avoiding unintentional doping. In Step 1, we inserted society, gender, age, education level, sport type (i.e., team or individual sports), and sport level (i.e., regional, state, national, international) as covariates in the prediction of the accuracy of avoiding unintentional doping. This step was to control for the potential confounding effects of society and the demographic backgrounds of athletes on doping-related behavioural outcomes.[24] Society was entered as two dummy-coded variables of Hong Kong (yes/no) and Australia (yes/no). In Step 2, implicit doping attitude served as the independent variable to examine the hypothesised negative relationship between implicit doping attitude and the accuracy of avoiding unintentional doping. Our hypothesis would be supported when the standardised beta coefficient of implicit doping attitude was statistically significant ($p < .05$) and negative (also indicated by the 95% confidence intervals of the unstandardised beta $\beta < 0$) in Step 2.

3. Results

No systematic pattern of missing data was detected (<1% for D-score, <2% for gender and <3% for sport type). Little’s MCAR test indicated that the missing data values were

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completely at random (χ^2 [1] = 1.28, p =.26). The assumption of collinearity was met for all variables (VIF < 1.5). Shapiro-Wilk's tests indicated that the scores of implicit doping attitude (p < .01) and the accuracy of avoiding unintentional doping (p < .01) were not normally distributed. However, we applied a hierarchical linear regression for the main analysis as in the previous study of implicit doping attitude[11] because our large sample size would protect the estimation of parameter estimates and standard errors from the violation of normality[25, 26], and the Central Limit Theorem assured that the regression coefficients were normally distributed on average.[27] Pearson correlations between the variables, means and standard deviations of the variables are displayed in Table 1.

INSERT TABLE 1 HERE

Hierarchical linear regression showed that implicit doping attitude (β = –.09, t [641] = -2.35, p = .02) was a significant and negative predictor of the accuracy in avoiding unintentional doping in Step 2 (R^2 = .029, F [1, 641] = 2.39, p = .02), after controlling for the effects of society, gender, age, education level, sport type, and sport level in Step 1 (R^2 = .021, F [7, 642] = 1.93, p = .06). Parameter estimates, confidence intervals, and effect sizes of the hierarchical linear regression are displayed in Table 2.

INSERT TABLE 2 HERE

4. Discussion

The aim of our study was to examine the extent to which athletes' implicit doping attitude is associated with their accuracy in avoiding unintentional doping by screening out beverages with banned performance-enhancing substances. The current findings supported our hypotheses that (1) athletes' implicit doping attitude towards doping was significantly and negatively associated with their accuracy in avoiding unintentional doping, and (2) this negative association was consistent across Hong Kong, Australia, and the United Kingdom.

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Implicit doping attitude is significantly and negatively linked to the accuracy of avoiding unintentional doping when controlling for the effects of society, gender, age, education level, sport type and sport level. Our finding is in agreement with previous research findings on the relationship between athletes' doping attitude and their other behavioural patterns of the avoidance of unintentional doping.[11] This indicates that athletes with a higher implicit attitude towards doping might be less efficient in avoiding unintentional doping, as they were less accurate in the task of screening out products that may contain illicit substances. Such findings may have important implications for future anti-doping initiatives to shift towards dual-processing models (explicit and implicit) as athletes' unconscious beliefs, values and attitude may implicate their subsequent risk of unintentional doping. However, it is important to point out that our proposed model with implicit doping attitude as the sole predictor only explained approximately 3% variance of the accuracy of avoiding unintentional doping. Although the effect size is comparable to that of the previous study of Chan and colleagues[11] (i.e., implicit doping attitude explained .70% to 8.1% variance of various behavioural outcomes of the avoidance of unintentional doping, the effect size is considered to be small. Future studies should revisit the predictive power of implicit doping attitude in relation to that of other psychological variables, and to examine its direct or indirect relationship with athletes' clean sport behaviours.

Our diverse sample from Hong Kong, Australia and the United Kingdom also addressed the limitation of previous studies on implicit doping attitude. We were able to show that the effects of society, gender, age, education level, sport type, and sport level did not significantly ($p > .05$) link to athletes' accuracy of the avoidance of unintentional doping. Moreover, since previous implicit doping studies were conducted predominately in Western countries, our diverse sample strengthened the generalisability of our findings to athletes from different societies or geographical regions.[28] Although athletes with different geographical backgrounds may carry different perspectives in striving for success in sport[29], athletes' accuracy of the

avoidance of unintentional doping appears to be a universal construct that is relevant to athletes in multiple societies. Future studies are encouraged to explore the possibility of developing an international training programme for enhancing athletes' accuracy of the avoidance of unintentional doping.

Another novelty of our study was the introduction of a new assessment tool in evaluating athletes' accuracy in avoiding unintentional doping. Previous studies that investigated athletes' avoidance of unintentional doping adopted the lollipop protocol, in which participants self-reported their behaviour of "not-taking", "not-eating", and "reading the ingredient list" towards a given lollipop of an unfamiliar brand.[11, 24] Although the responses were verified by research associates, the responses of "reading the ingredient list" could be subjected to social desirability, and could not reveal the extent to which athletes accurately and efficiently identified the presence of banned performance enhancing substances in the ingredient lists while avoiding unintentional doping by refusing to consume the food product. Our canned beverage sorting task was able to fill this methodological gap by objectively testing athletes' accuracy in avoiding unintentional doping in a real-time experiment where the accuracy was recorded objectively by the computer system. This task is more likely to simulate real-life situations where athletes have to be consistently vigilant in screening out food, drinks, or medication that may contain banned performance-enhancing substances.[1, 7, 30, 31] The negative association between athletes' implicit doping attitude and their accuracy of avoiding unintentional doping could provide initial evidence regarding the concurrent validity of the canned beverage sorting task. As the canned beverage sorting task is a highly accessible and convenient tool that can be executed on athletes' smartphones, the task may offer a potential solution to the World Anti-Doping Agency's strategic plan of social science research[32] in developing an assessment for evaluating athletes' behavioural patterns of unintentional doping. Such a tool can serve as an assessment for future doping initiatives, and the World Anti-Doping Agency to objectively

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measure behaviours relating to the avoidance of unintentional doping. Unlike the current literature, which heavily relies on athletes' self-reported behaviours, this tool could provide with a more accurate and realistic assessment of athletes' practices regarding banned substances. Through supplementing existing research with this novel measurement tool, it could allow the World Anti-Doping Agency and research to develop more targeted interventions, educational practice and policy, ultimately addressing a wholistic anti-doping system.

In sum, our study utilised two objective experimental tasks in evaluating implicit doping attitude and the accuracy of avoiding unintentional doping among athletes from three societies. Despite the strengths and unique findings of our study, we must acknowledge a number of limitations. First, our canned beverage sorting task was only designed to capture athletes' accuracy and vigilance in detecting and avoiding beverages with the presence of banned performance-enhancing substances as shown in the ingredient table of the beverage. However, athletes' knowledge could be a variable in determining whether the beverage contained banned-substances, so we did not use technical names of banned-performance substances (i.e., Adrafinil, Cropropamide). Instead, we only used categories as these are more widely known and detectable by general athletes. These categories were similarly used in the IAT task, hence ensuring that the athlete participants were equipped with the knowledge of these banned-substances. Additionally, it is equally important to note that unintentional doping could also happen when drugs or dietary supplements are contaminated with prohibited substances due to poor manufacturing practices, making the action of checking the ingredient table misleading or insufficient.[2, 3] Thus, it is important that athletes consult sports medical professionals (e.g., team physicians or sports dietitians) before they consume any new drugs or dietary supplements, and check the ingredients of any consumable energy products they are not familiar with. Second, apart from doping attitude, there are several psychological factors such as motivations, social cognitive beliefs, self-control, and intention that could potentially be related to athletes'

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behavioural patterns of the avoidance of unintentional doping.[1, 33, 34] Future studies may examine if the salience of implicit doping attitude would hold when taking these psychological factors into account.[35] Third, whilst implicit association tests are used to measure automatic or unconscious attitudes, the test is not impervious to practice effects or response familiarity or stimuli familarity.[36, 37] To minimize these effects, however, our study design incorporated randomisation of testing blocks in an attempt to counterbalance order effects or task familiarity effects. Forth, our sample consisted of athletes whose competitive levels were primarily at regional or state level, and there were fewer athletes (23.4%) who competed at national or international level. The experience of anti-doping/ doping control, and the knowledge of banned performance-enhancing substance might be considerably better among athletes at higher competitive level, which might affect their interpretation and responses to our key study measures concerning about doping. Although we statistically controlled for the explained variance of sport level in the analysis, future studies should consider athletes' prior experience of anti-doping education, doping control, and the knowledge of banned performance-enhancing substances in the sampling, measurement, and statistical analysis. Finally, under a cross-sectional study design, the assessment of the study variables was only conducted on a single occasion, and no follow-up measures or manipulation of doping attitude/ unintentional doping behaviour was done. Therefore, our findings were unable to offer evidence regarding the causal or temporal relationships between implicit doping attitude and athletes' avoidance of unintentional doping. Future research should consider adopting a longitudinal study design to examine the predictive power of implicit doping attitude on athletes' unintentional doping behaviours, and investigate behavioural strategies to enhance athletes' attitude and behavioural patterns towards the avoidance of unintentional doping. Interventions with applications of theory-driven behavioural change strategies (e.g., provide information on consequences, prompt self-monitoring of behaviour and action planning)[38] have been shown to be useful for fostering certain health behaviours (e.g., physical activity); as such, further studies can explore

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the feasibility of applying these strategies in the anti-doping context and see if they can facilitate athletes' adaptive psychological and behavioural patterns in avoiding unintentional doping.

5. Conclusion

The present study investigated the extent to which implicit doping attitude correlated with athletes' avoidance of unintentional doping. The findings from Hong Kong, Australia, and the United Kingdom supported the hypotheses that athletes' implicit doping attitude was significantly and negatively associated with their accuracy in avoiding unintentional doping, and that this association was maintained across societies. Athletes who hold positive implicit doping attitude were less accurate in detecting and avoiding energy beverages with the presence of banned performance-enhancing substances.

Practical implications

- The smartphone version of the implicit association test of doping attitude appears to be a convenient assessment tool for measuring athletes' implicit attitude towards doping.
- The novel canned beverage sorting task may also provide an objective measure of athletes' accuracy in identifying energy beverages containing banned performance-enhancing substances.
- Both assessments appear to be robust and applicable to athletes from different societies, so future anti-doping educational programmes or interventions may consider using them as objective outcome measures, as opposed to the current literature which heavily relies on athletes' self-report.

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Table 1

Correlation matrix, mean, and standard deviation of the study variables.

	1	2
Predictors		
1. Implicit Doping Attitude	1	
Behavioural Outcome		
2. Accuracy of Avoiding Unintentional Doping	-.10**	1
Control Variables		
3. Society	-.02	.03
4. Gender	-.06	-.03
5. Age	-.01	-.06
6. Education Level	-.00	-.04
7. Type of Sport	.01	-.01
8. Sport Level	-.02	.06
Mean	-.38	27.83
SD	.50	4.93

**$p < .01$ at 2-tailed.

Note. Society (1 = Hong Kong, 3 = Australia, 4 = The United Kingdom); Gender (1 = male, 2 = female); Education (1 = secondary school, 2 = post-school training college, 3 = university undergraduate, 4 = university postgraduate); Type of sport (0 = team sport, 1 = individual sport); Sport level (1 = regional level, 2 = state level, 3 = national level, 4 = international level, 5 = world top 20).

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Table 2

Results of hierarchical linear multiple regression models

Step	Independent Variables	β	(95% CI of β)	R^2	ΔR^2
1	Hong Kong (yes/no)	-.06	-1.72 to .34	.02	.02
	Australia (yes/no)	.09	-.01 to 1.93		
	Gender (male/female)	-.02	-.97 to .57		
	Age	-.06	-.09 to .01		
	Education	-.03	-.55 to .27		
	Sport Type	.02	-.58 to .97		
	Sport Level	.03	-.32 to .71		
2	Implicit Doping Attitude	-.09*	-1.64 to -.15	.03*	.01*

* $p < .05$

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Appendix 1

ESTABLISHED SINCE 2014
RED STAR
ENERGY DRINK
Nutrition Facts

Confirmation of Ethical Compliance

This study was approved by the Institutional Review Board of Hong Kong Hospital Authority Western Cluster. The authors have ensured that the study was carried out in accordance with The Code of Ethics of the World Medical Association.

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CRediT Author Statement

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Declarations of Interest

None

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